

Grounded Agent Reputation

Facts about AI agents and their reviewers, proven from Ethereum into Creditcoin

BUIDL CTC 2026 Fall · Track: AI · Creditcoin CC3 Testnet + Attestcoin

The problem (Ethereum mainnet, measured 29 Aug 2026)

- 346 of 367 rated agents have exactly **one** reviewer; one EOA wrote **225** reviews for 195 agents
- 16 of 105 reviewers **own agents** and wrote **59%** of all feedback
- Last 60 days: **14,771** registrations, **83%** from owners holding ≥ 10 agents; 8,136 agents minted in batch txs
- Independent study: 73.5% of Ethereum reviewers show coordinated Sybil behaviour (arXiv 2606.26028)
- The spec's answer: "aggregation will happen off-chain" → trust an aggregator (RNWY, MainStreet, Kleros)

The idea: facts, not scores

- `GroundedFacts` (Creditcoin) admits data **only** through Attestcoin proofs of Ethereum txs
- Facts per agent: reviewers proven real, reviewer seniority (oldest tx, activity months), gaps in review indices, negatives, clone density by owner **and by registrant**, same-tx siblings, reviewer-owns-agents
- No weights. The consumer passes `minAge`, `minDepth`, `k`, `c`; the contract answers numbers anyone can recompute (`verify.ts`)

Attestcoin depth (all measured)

Capability	Evidence
Mainnet (chainKey 3) + Sepolia (chainKey 1) in one contract	<code>attested-height/3</code> , <code>ChainInfo</code> <code>is_height_attested</code> = true
Proof of a 2.5-year-old tx verified on-chain	604 roots, <code>verify</code> = true, 414,624 gas
Registry event proof	<code>NewFeedback</code> 117,971 gas; contract decode+storage 259k
Dedup via <code>calculateTxIndex</code> ; official <code>EvmV1Decoder</code> ; registry counter <code>feedbackIndex</code> as completeness oracle	tests + testnet txs

Money follows facts (no reserve, no oracle)

- `AgentHireEscrow` : $\text{premium} = \text{base} + (\text{max} - \text{base}) \cdot \text{risk}$; risk falls with senior reviewers and rises with clone density; premium paid **to the agent** (earn real reviewers)
- `CoverageBounty` : pays whoever submits proofs that **change the consumer's decision** — evidence that hurts an agent pays the same as evidence that helps

The agent: GroundedScout (4 decisions, all logged)

- **R1 Targeting** — open bounties first, then most-reviewed agents in the last 7 days
- **R2 Two-way evidence** — senior reviewers (helps) and higher indices / negatives / clone siblings / reviewer-owns-agents (hurts); 40% of gas reserved for helps
- **R3 Timing** — skip proofs already admitted (`txSeen`); prove only when bounty $\geq$ cost
- **R4 Consumer** — hire through the escrow when its own thresholds pass, else fund a bounty
- No LLM in the fact path; the contract recomputes everything from bytes

Demo (mainnet data, Creditcoin txs)

- Agent **34135**: 3 senior reviewers → premium 2%
- Agent **50283**: 1 reviewer who owns **43 agents**, 5 clone siblings proven → premium 20%
- Tx hashes on both chains in the video; every number reproducible from the README commands

Limits we state upfront

- Ethereum-side registries only; Base ($\approx 139\times$ more activity) is out of Attestcoin's reach today
- Newest unsubmitted reviews are undetectable (only gaps below the highest proven index)
- Clone density is a lower bound; aged wallets can be bought; legit multi-agent operators look like farms
- No consumer contract on Creditcoin yet — the escrow is an example consumer
- Attestcoin moves trust from RPC/indexers to Creditcoin's bonded attestor set; it does not remove it

Prior art and where we differ

	RNWY / AgentRadar	Kleros oracle	MainStreet	Grounded
Signal	wallet age, clones	jury decisions	feedback + x402	proven facts
Where	API	registry (Sepolia)	registry (Base)	Creditcoin contract
Verifiable by a contract	no	trust router	trust publisher	yes, from proofs
Score	yes	yes	yes	no — thresholds

Team & links

- Dien — builder (Veritas UH19 winner: dynamic fee / DRS pattern reused here without the insurance reserve)
- Repo: `<GITHUB_URL>` · Contracts: `<FACTS>` , `<ESCROW>` , `<BOUNTY>` · Video: `<VIDEO_URL>`
- Dossier with verification commands: `docs/evaluation-dossier.md`